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Variance Partitioning of Climate and Topographic Controls on High-Andean Vegetatio Dynamics: A Spatio-Temporal Machine-Learning Analysis of the Peruvian Altiplano (2015–2025)

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Abstract:	High-Andean puna ecosystems of the Peruvian Altiplano are among the most climate-sensitive grasslands on Earth, yet the relative contribution of topographic versus climatic drivers to their vegetation dynamics remains largely unquantified. Pooled space–time models often report optimistic performance because spatial gradients and temporal autocorrelation inflate cross-validated scores; here we explicitly partition NDVI variance into a between-pixel (spatial) and a within-pixel (temporal) component before modeling. Using MODIS NDVI (MOD13Q1), CHIRPS v2.0 precipitation, MODIS LST and SRTM terrain data at ~1 km resolution for the Peruvian Altiplano (2015–2025), restricted to elevations above 3000 m (175,780 pixels, 16.7 million observations), we find that 69.8% of total NDVI variance is spatial and 30.5% is temporal. A nested sequence of random-forest experiments reveals that topography and latitude account for the bulk of the spatial signal (R2 = 0.785), while climate anomalies explain 14.7 percentage points of within-pixel variance beyond a pure seasonal baseline (R2 from 0.544 to 0.691). Thermal anomalies dominate the hydric signal, and their interaction with elevation raises performance to R2 = 0.750; partial-dependence analysis shows the vegetation response to thermal stress weakens with elevation. A pooled model validated with leave-one-year-out and spatial 1block cross-validation reaches R2 = 0.823 without the leakage inflation typical of naive space–time models. These results provide the first explicit variance partition of high-Andean vegetation dynamics, establish the primacy of the spatial gradient over interannual climate, and identify the elevation-modulated thermal signal as the dominant short-term climate control.
Suggested Reviewers:	Hanna Meyer hanna.meyer@uni-muenster.de Dr. Hanna Meyer is a leading expert in geospatial machine learning and the development of spatial cross-validation strategies. Her work directly addresses spatial autocorrelation and overfitting in environmental models, which is highly relevant to our manuscript's core methodology of partitioning NDVI variance into spatial (69.8%) and temporal (30.2%) components using spatial block cross-validation.
	Wim Buytaert w.buytaert@imperial.ac.uk Prof. Wim Buytaert is an international authority on the hydrology and climate change vulnerability of High-Andean ecosystems, specifically puna grasslands and wetlands (bofedales). His deep understanding of high-altitude environmental gradients and their ecological responses is essential to evaluate our findings regarding the elevation-modulated climate sensitivity of the Altiplano vegetation.
	Mathias Vuille mvuille@albany.edu Prof. Mathias Vuille is a leading climatologist specializing in climate change, temperature dynamics, and convective precipitation patterns across the tropical Andes. His expertise is crucial for assessing our analysis of temperature and precipitation

	anomalies (using ERA5 and LST) and our specific findings regarding the modulation of thermal stress along the elevational gradient of the Altiplano.
	Claudia Canedo-Rosso claudia.canedo_rosso@nateko.lu.se Dr. Claudia Canedo-Rosso has published extensively on precipitation variability, droughts, and their ecological and agricultural impacts specifically within the Altiplano region. Her local geographical knowledge and hydro-climatic expertise make her exceptionally qualified to evaluate our manuscript's climate-anomaly experimental design and our findings regarding the performance of multi-scale SPI indices.
	Fred Torres Cruz ftorres@unap.edu.pe Ing. Fred Torres Cruz is an expert in statistical computing, machine learning, and satellite image processing at the Universidad Nacional del Altiplano. He has developed specialized software for time-series anomaly detection (ANOMALIAPRO) and satellite-based feature tracking (GLOBALTRACE). His dual expertise in spatial analysis, time-series modeling, and machine learning makes him highly qualified to evaluate our spatio-temporal variance partitioning framework on the Altiplano.
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Cover letter

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1 August 2026

Dear Editor-in-Chief,

I am pleased to submit the manuscript entitled "Variance Partitioning of Climate and Topographic Controls on High-Andean Vegetation Dynamics: A Spatio-Temporal Machine-Learning Analysis of the Peruvian Altiplano (2015-2025)" for consideration for publication in Global and Planetary Change.

The study provides the first explicit decomposition of NDVI variance in the Peruvian Altiplano into spatial and temporal components, showing that 69.8% of the variance is spatial (between-pixel) and 30.5% is temporal (within-pixel). Using a nested sequence of random-forest experiments with leakage-robust validation (leave-one-year-out and spatial-block cross-validation), we show that latitude and elevation dominate the spatial gradient, that temperature anomalies are a stronger short-term climate control than precipitation anomalies and SPI drought indicators, and that the vegetation response to thermal stress weakens with elevation. The results directly expose the autocorrelation inflation of naive space-time models and provide a transferable analytical framework for high-Andean vegetation monitoring. We believe the study fits the journal's focus on global change processes and their regional manifestations, particularly in climate-sensitive mountain ecosystems such as the Andean Altiplano under accelerated warming.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All source data (MODIS, CHIRPS, SRTM) are publicly available, and the full analysis pipeline is available from the corresponding author on reasonable request. The author declares no competing interests. During the preparation of this work the author used large-language-model tools for drafting and language-editing assistance; the author reviewed and edited the content and takes full responsibility for the published material.

I confirm that this manuscript has been read and approved by its sole author and that no other persons satisfied the criteria for authorship. Thank you for your consideration.

Sincerely,

Noe Ulises Machaca Chambilla

Highlights

- 69.8% of Altiplano NDVI variance is spatial, only 30.5% temporal
- Latitude, not elevation, dominates the spatial gradient of puna greenness
- Temperature anomalies add +14.7 pp over seasonality; SPI adds none
- Thermal sensitivity of NDVI weakens with elevation across the puna
- Leakage-robust CV yields R2=0.82, exposing naive-model inflation (0.98)

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Variance Partitioning of Climate and Topographic Controls on High-Andean Vegetation Dynamics: A Spatio-Temporal Machine-Learning Analysis of the Peruvian Altiplano (2015–2025)

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Abstract

High-Andean puna ecosystems of the Peruvian Altiplano are among the most climate-sensitive grasslands on Earth, yet the relative contribution of topographic versus climatic drivers to their vegetation dynamics remains largely unquantified. Pooled space–time models often report optimistic performance because spatial gradients and temporal autocorrelation inflate cross-validated scores; here we explicitly partition NDVI variance into a between-pixel (spatial) and a within-pixel (temporal) component before modeling. Using MODIS NDVI (MOD13Q1), CHIRPS v2.0 precipitation, MODIS LST and SRTM terrain data at ∼1 km resolution for the Peruvian Altiplano (2015–2025), restricted to elevations above 3000 m (175,780 pixels, 16.7 million observations), we find that 69.8% of total NDVI variance is spatial and 30.5% is temporal. A nested sequence of random-forest experiments reveals that topography and latitude account for the bulk of the spatial signal ($R^2 = 0.785$), while climate anomalies explain 14.7 percentage points of within-pixel variance beyond a pure seasonal baseline (R^2 from 0.544 to 0.691). Thermal anomalies dominate the hydric signal, and their interaction with elevation raises performance to $R^2 = 0.750$; partial-dependence analysis shows the vegetation response to thermal stress weakens with elevation. A pooled model validated with leave-one-year-out and spatial

block cross-validation reaches $R^2 = 0.823$ without the leakage inflation typical of naive space–time models. These results provide the first explicit variance partition of high-Andean vegetation dynamics, establish the primacy of the spatial gradient over interannual climate, and identify the elevation-modulated thermal signal as the dominant short-term climate control.
Keywords: high-Andean vegetation; variance partitioning; random forest; spatio-temporal modeling; MODIS NDVI; Peruvian Altiplano

1 Introduction

1.1 Elevation gradients and high-Andean vegetation

The Andean Altiplano is a high plateau of approximately 200,000 km^2 surrounded by the Western and Eastern Cordilleras, most of it above 3800 m above sea level (Garreaud 2009). Its puna vegetation is dominated by tussock grasses, cushion plants and scattered wetlands (bofedales) that sustain pastoral economies and constitute key carbon and water stores (Buytaert et al. 2011). Because temperature decreases with elevation and precipitation is delivered by a strong seasonal cycle (Canedo-Rosso et al. 2018), the altitudinal gradient acts as a natural laboratory for studying how vegetation responds to climate forcing (Pepin et al. 2015; Vuille et al. 2015). The greenness signal captured by the Normalized Difference Vegetation Index (NDVI) is a widely used proxy of photosynthetic activity and canopy condition (Tucker 1979), and satellite time series now permit tracking its spatio-temporal variability across the entire plateau (Tamiminia et al. 2020; Greifeneder et al. 2021).

1.2 The paradox of space–time predictability

A first pooled analysis of NDVI in the study domain (Appendix A) achieved a very high out-of-sample performance ($R^2 = 0.98$), largely because the model captured the persistent spatial gradient and the strong temporal autocorrelation of the series (lag-1 autocorrelation of the pooled residuals of 95.4%). Such scores are deceptive: when the same observations appear in both training and test folds through temporal or spatial proximity, the apparent predictive skill conflates memorization with generalization (Ogle et al. 2015; Roberts et al. 2017). In ecological time series, temporal autocorrelation alone can inflate naive R^2 estimates severalfold (Ogle et al. 2015), and spatial blocking is required to produce honest estimates of generalization error (Hodges and Reich 2010; Roberts et al. 2017; Meyer and Pebesma 2022). The diagnostic therefore reveals not a well-understood system but a modeling problem: the variance of the response is dominated by components that naive pooled models attribute to “fit” without distinguishing between spatial, seasonal and interannual sources.

1.3 Variance partitioning as a solution

Variance-partitioning approaches decompose the total variability of a response into orthogonal spatial and temporal components (Borcard et al. 1992; Meyer, Reudenbach, et al. 2019), making explicit how much signal each modeling family can in principle recover. Applied to NDVI, this decomposition answers three questions that pooled models blur: (i) how much of the total variability is spatial (between pixels) and how much is temporal (within pixels); (ii) which topographic variables explain the spatial component; and (iii) which climate variables explain the temporal component after removing the seasonal cycle. Machine-learning regressors such as random forests (Breiman 2001) are well suited to this task because they capture nonlinear and interaction effects without parametric assumptions, while permutation or impurity-based importance and Shapley-based explanations (Meyer, Reudenbach, et al. 2019; Lundberg and Lee 2017) provide interpretability.

1.4 Objectives

The general objective is to quantify the spatio-temporal drivers of vegetation dynamics in the Peruvian Altiplano (2015–2025) through a variance-partitioning framework with machine learning. Specific objectives are: (1) partition the total NDVI variance into between-pixel (spatial) and within-pixel (temporal) components; (2) quantify the spatial gradient explained by topography and geography; (3) measure the interannual climate signal (temperature and precipitation anomalies and drought indices) beyond a pure seasonal baseline; (4) test whether the climate response is modulated by elevation; and (5) assess the generalization of a pooled model under leave-one-year-out and spatial-block cross-validation.

2 Study Area and Data

2.1 Study area

The study area covers the Peruvian segment of the Altiplano (approximately 72.0°–68.5°W, 18.5°–13.0°S). Following the variance-partitioning rationale, the domain was restricted to elevations above 3000 m to exclude the eastern lowland forests of the Amazon basin, which dominated the raw bounding box (∼30% of pixels below 3000 m) and would otherwise confound the elevational signal. The restricted domain retains 175,780 pixels (70.2% of the bounding box) on a 0.00833° (∼1 km) grid, of which 90.3% lie above 3800 m; median elevation is 4201 m with an interquartile range of 3910–4530 m and a maximum of 6331 m. Figure 1 shows the elevation mask and the mean NDVI field.

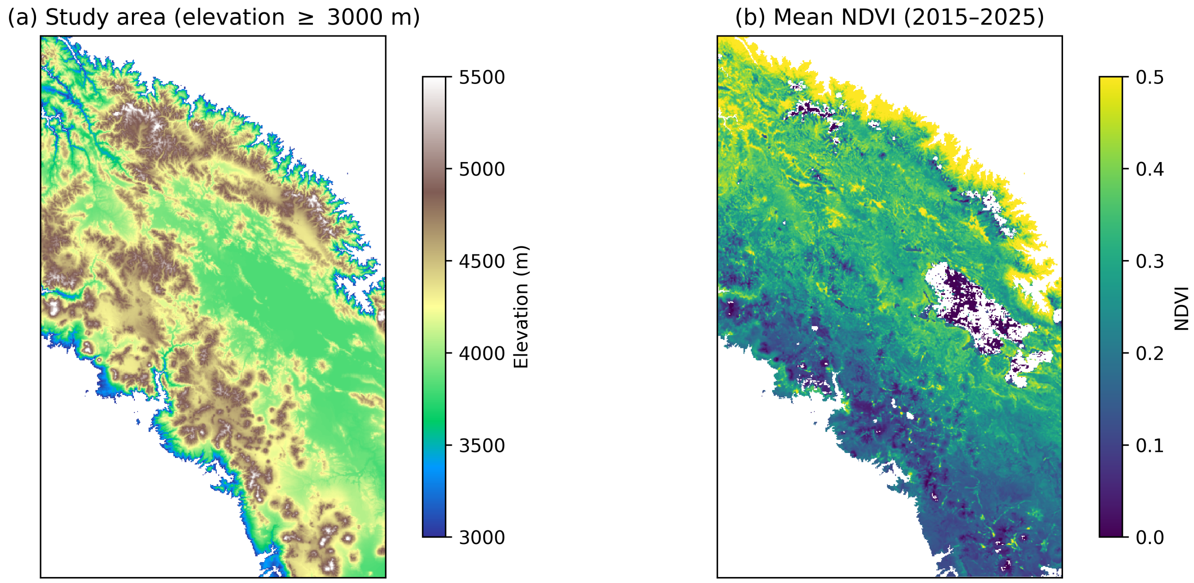

Figure 1: Study area restricted to elevations above 3000 m. (a) Elevation from SRTM (m) with the Andes cordilleras delimiting the plateau; (b) mean NDVI (2015–2025) showing the strong gradient from dry low-altitude puna to the productive northwestern sector.

2.2 Data

We used three data sources. First, vegetation greenness from the MODIS MOD13Q1 v6.1 product (16-day composites at 250 m), aggregated to monthly means through the Google Earth Engine (Tamiminia et al. 2020); the series spans 2015–2025 (132 months). Second, climate from CHIRPS v2.0 precipitation (mm month^{-1}) and MODIS land surface temperature (MOD11A2, 8-day products averaged to monthly means in °C). Third, topography from SRTM at 3 arc-seconds: elevation, slope, aspect, and Euclidean distance to the nearest lake. All variables were aligned to a common 0.00833° ($\sim$1 km) grid. The complete dataset contains 16,666,109 pixel–date observations after the elevation restriction.

Drought conditions were characterized with the Standardized Precipitation Index at 3, 6 and 12-month scales (McKee et al. 1993). Climate anomalies were computed as deviations of each monthly value from its long-term mean within the series (temperature anomalies, *LST_anom*, and precipitation anomalies, *precip_anom*), removing the seasonal climatology so that the temporal models isolate interannual variability rather than the annual cycle.

Table 1 summarizes the key variables. The mean NDVI of 0.27 with a standard deviation of 0.13 reflects the sparse vegetation of the puna; precipitation is strongly right-skewed (mean 43.6 mm per month, median 22.6 mm), and temperature averages 25.0°C at the pixel-level seasonal scale. The mean annual cycle (Figure 2) shows the wet-season greening between December and March and the dry-season senescence between June and September, with a secondary precipitation peak and the well-known austral-summer maximum of NDVI.

Table 1: Descriptive statistics of the restricted study domain (n = 175,780 pixels; 16,666,109 pixel–date observations, 2015–2025). Precipitation and temperature are monthly means; NDVI is the monthly MOD13Q1 mean.

Variable	Mean	SD	Median	p5	p95	Min–Max
NDVI	0.270	0.134	0.243	0.106	0.538	−0.132 to 0.924
Precipitation (mm)	43.6	50.8	22.6	0.0	148.7	0.0 to 754.0
Temperature (°C)	25.0	6.8	24.0	15.2	37.2	−0.4 to 53.3
Elevation (m)	4211	395	4202	3517	4821	3000 to 6331

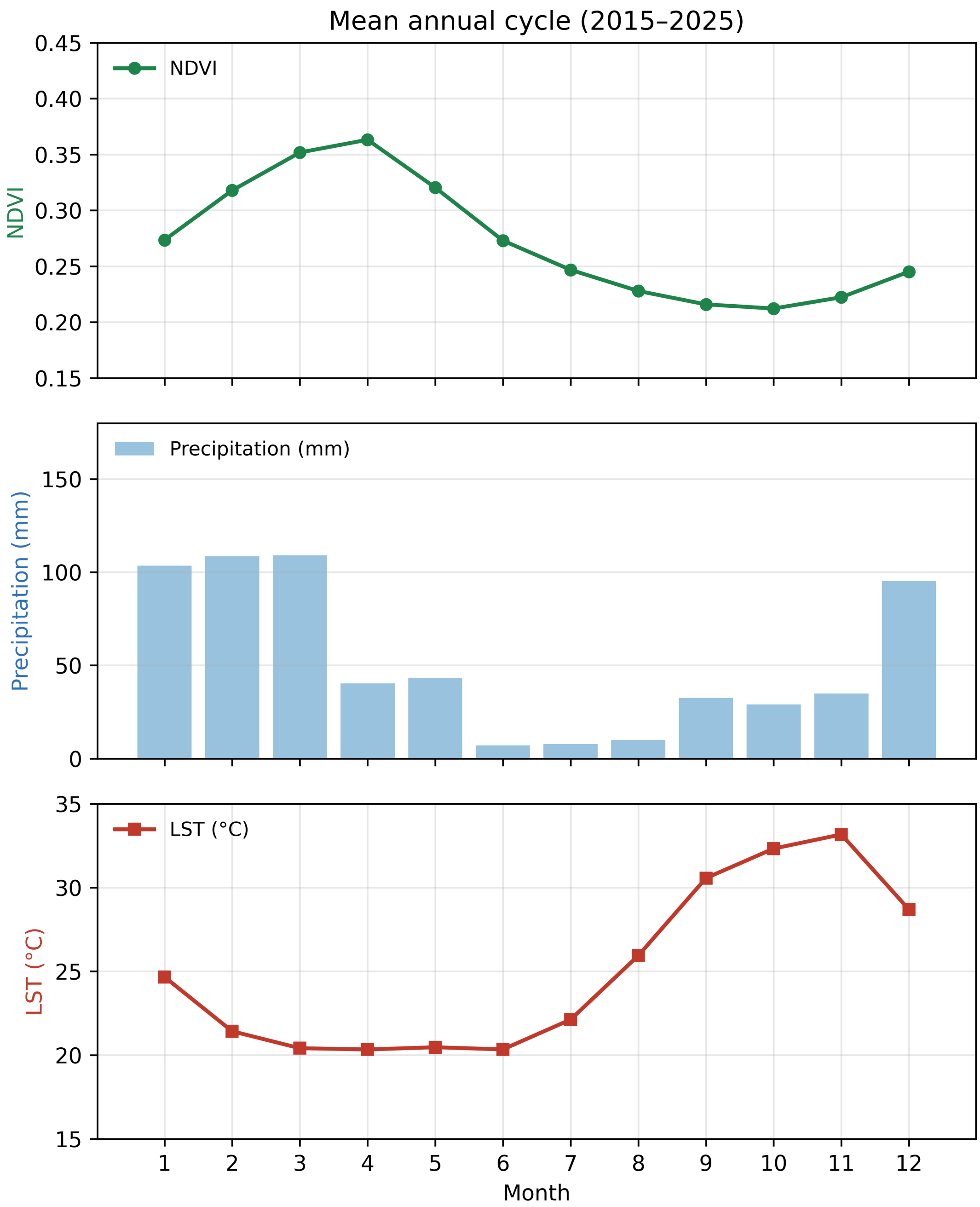

Figure 2: Mean annual cycle (2015–2025) of NDVI, precipitation and temperature across the restricted domain. NDVI peaks in the wet season (February) and reaches a minimum in the dry season (September).

3 Methods

3.1 Variance decomposition

Let y_{pt} be the NDVI of pixel p at time t, μ_p the temporal mean of pixel p, and μ the global mean. The total variance is decomposed into a between-pixel (spatial) component and a within-pixel (temporal) component (Borcard et al. 1992):

$$\underbrace{\frac{1}{N}\sum_p\sum_t (y_{pt}-\mu_p)^2}_{\text{within-pixel (temporal)}} + \underbrace{\frac{1}{N}\sum_p n_p(\mu_p-\mu)^2}_{\text{between-pixel (spatial)}} = \frac{1}{N}\sum_p\sum_t (y_{pt}-\mu)^2, \tag{1}$$

where N is the total number of observations and n_p the number of dates of pixel p. The fractions $\sigma^2_{\text{between}}/\sigma^2_{\text{total}}$ and $\sigma^2_{\text{within}}/\sigma^2_{\text{total}}$ express how much of the total variability is attributable to spatial gradients versus temporal dynamics, respectively. We then model each component with a separate random forest to attribute it to specific predictors.

3.2 Experimental design

A nested sequence of random-forest experiments isolates the contribution of each source of variance. All temporal models predict the *centered* NDVI (deviation of each observation from the pixel's temporal mean, $y_{pt} - \mu_p$), so that the spatial gradient is removed by construction and the experiments target only the within-pixel variance. Table 2 defines the models.

Table 2: Experimental design. All temporal models (T0, T1, I) predict the pixel-centered NDVI; S predicts the pixel temporal mean NDVI.

Model	Response	Predictors
S (RF / OLS)	μ_p (temporal mean)	elevation, y, x, slope, aspect, dist_lake
T0	$y_{pt} - \mu_p$	sin_month, cos_month (seasonality)
T1a	$y_{pt} - \mu_p$	seasonality + precip_anom, LST_anom
T1b	$y_{pt} - \mu_p$	seasonality + SPI3, SPI6, SPI12
T1c	$y_{pt} - \mu_p$	seasonality + precip_anom, LST_anom, SPI3, SPI6, SPI12
I	$y_{pt} - \mu_p$	seasonality + precip_anom, LST_anom, elevation, precip_anom×elevation, LST_anom×elevation
C (pooled)	y_{pt} (raw NDVI)	all geographic, topographic, seasonal and climate features

3.3 Models and validation

Random forests (Breiman 2001) were configured with 500 trees, a maximum depth of 15, a minimum leaf size of 5 and bootstrap sampling with subsampling to control memory (results are robust across configurations). Linear regression was additionally fitted for the spatial model S to provide a parametric baseline. Performance was measured with R^2, RMSE and MAE under three validation schemes that match the structure of each experiment:

- **10-fold cross-validation** for the spatial model S, applied at the pixel level;

- **leave-one-year-out** (LOYO) for the temporal models T0, T1 and I, so that a full year of observations is held out and the test data are never temporally adjacent to the training data;

- for the pooled model C, both **leave-one-year-out** and **spatial block cross-validation** (10-fold *GroupKFold* on 0.0833° blocks), which together guard against temporal and spatial leakage (Roberts et al. 2017; Meyer and Pebesma 2022).

Feature importance was quantified with impurity-based importance for all random forests (Meyer, Reudenbach, et al. 2019) and with SHAP values (Lundberg and Lee 2017) for the main models (T1a and C), computed on a random subsample of 50,000 rows with a compact 30-tree forest. Conditional partial dependence (Meyer, Reudenbach, et al. 2019) of the interaction model I was computed to characterize how the response to temperature and precipitation anomalies changes across the elevational gradient (at the 10th, 50th and 90th percentiles of elevation).

4 Results

4.1 Global variance partition

The total NDVI variance of the restricted domain is 0.0178. Of this, 69.8% is between-pixel (spatial) variance and 30.5% is within-pixel (temporal) variance (Figure 3a). The spatial gradient therefore dominates the variability of high-Andean greenness by a factor of 2.3, even within a domain that is already restricted to elevations above 3000 m.

Modeling the between-pixel component (S, $R^2 = 0.785$) attributes it almost entirely to geographic and topographic predictors: latitude contributes 43.4% of the total variance, elevation 20.0%, longitude 11.7% and the remaining terrain attributes (slope, aspect, distance to lake) 3.3%, with 21.5% unexplained (Figure 3b). The dominant role of latitude reflects the northwest–southeast moisture gradient of the plateau, which is only partially captured by the discrete elevation field.

Modeling the within-pixel component (I, $R^2 = 0.750$ of the temporal variance) shows that seasonality explains 54.4 percentage points, interannual climate anomalies an additional 14.7 points, and the elevation modulation of the climate response a further 5.9 points; 25.0 points remain unexplained (Figure 3c). Taken together, the decomposition shows that the temporal dynamics are only partially seasonal: a substantial part of the interannual signal is a direct, elevation-modulated response to climate anomalies.

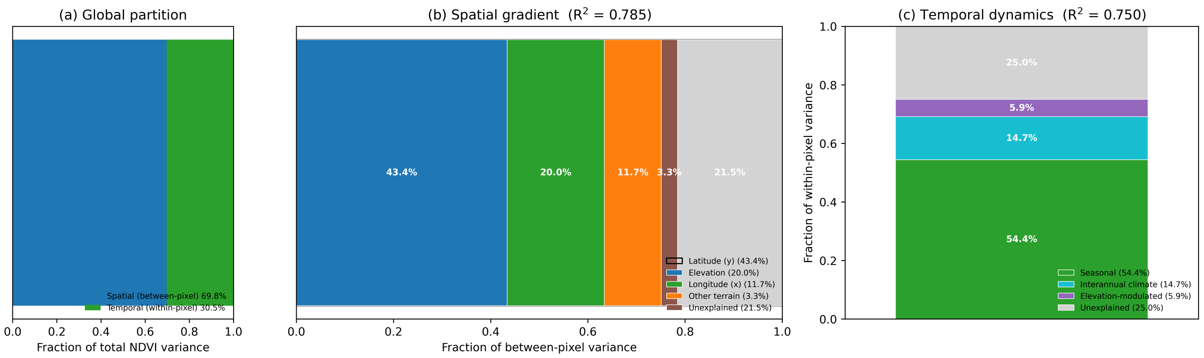

Figure 3: Global variance partition of NDVI in the Peruvian Altiplano (2015–2025). (a) Between-pixel (spatial) vs. within-pixel (temporal) fractions of total variance. (b) Attribution of the between-pixel variance to predictors ($R^2 = 0.785$). (c) Attribution of the within-pixel variance to seasonal, interannual climate and elevation-modulated components ($R^2 = 0.750$).

4.2 Spatial model S

The random forest on pixel temporal means reaches $R^2 = 0.785 \pm 0.041$ (RMSE $= 0.055$), against $R^2 = 0.706$ for ordinary least squares. The RF advantage confirms nonlinear structure in the elevational response, which is steep below roughly 4300 m and flattens toward the cold upper puna. Feature importance ranks latitude (0.553) far above elevation (0.255) and longitude (0.149), confirming that the north–south moisture gradient is the first-order spatial control of greenness in the Peruvian sector, with elevation as a secondary thermal control.

4.3 Temporal models T0, T1a, T1b and T1c

Table 3 reports all experiments. The seasonal baseline T0 reaches $R^2 = 0.544 \pm 0.044$ on centered NDVI: slightly more than half of the temporal variability is captured by the calendar cycle. Adding the temperature and precipitation anomalies (T1a) raises performance to $R^2 = 0.691 \pm 0.055$ (+14.7 percentage points), a large and systematic gain. The importance analysis attributes this gain primarily to thermal anomalies: *LST_anom* importance is 0.093 against 0.052 for *precip_anom*, and the seasonal harmonics still carry the largest share (0.842 combined). SHAP values confirm the asymmetry, with a mean absolute contribution of *LST_anom* of 0.0121 versus 0.0025 for *precip_anom*.

Replacing the anomalies by drought indices (T1b, SPI3/6/12) performs worse than the seasonal baseline ($R^2 = 0.517 \pm 0.173$). This counterintuitive negative result is methodologically informative rather than a null finding. First, the SPI is a precipitation-only index: it excludes temperature and evapotranspiration, and therefore carries no information on the thermal forcing that T1a identifies as dominant. Second, the three accumulation horizons (3, 6 and 12 months) are strongly collinear, which misleads the greedy feature splitting of the random forest. Third, the precipitation deficits summarized over multi-month windows add no signal beyond the calendar cycle at monthly resolution, and this absence is only made visible by the strict leave-one-year-out scheme; a random-split validation would have masked it. Combining both anomaly families (T1c) reaches $R^2 = 0.649 \pm 0.222$ but with high instability across years (one negative fold in 2015), consistent with the hydrological extremes of that year. The temperature anomaly is thus the single most informative interannual climate variable, and the precipitation signal is comparatively weak at the monthly temporal resolution of the analysis.

4.4 Interaction model I and partial dependence

Explicitly allowing the climate response to depend on elevation (I) improves performance to $R^2 = 0.750 \pm 0.056$, +5.9 percentage points over T1a. Impurity importance places the interaction *LST_anom*× *elevation* (0.039) above the direct effect of either anomaly, confirming that the thermal response is elevation-dependent. Conditional partial dependence (Figure 4) shows that the response to temperature anomalies is strongly negative at all elevations but steepens at low elevation: a thermal anomaly of +8°C reduces predicted centered NDVI by roughly 0.085 at 3830 m but only 0.036 at 4736 m. The precipitation response is weak and non-monotonic, consistent with the low importance of *precip_anom*. Its V- or W-shaped profile around zero anomalies must be interpreted as a numerical artifact, not as a physiological response: the zero-inflation of the dry season (May–August), when precipitation is structurally zero, concentrates a large density of observations at near-zero anomalies around which the random forest creates fine, locally unstable partitions. These translate into small non-physical wiggles in the marginal response (amplitude ±0.007 NDVI, an order of magnitude below the thermal swing) and break any point-wise reading of the curve. The robust hydric finding is therefore the overall flatness of the profile – a weak sensitivity of monthly NDVI to short-term hydric anomalies – in sharp contrast with the steep, monotonic slope of the thermal response (Figure 4, left). This implies that vegetation at the lower boundary of the domain is most sensitive to thermal stress, while the coldest, highest puna is buffered by temperature limitation.

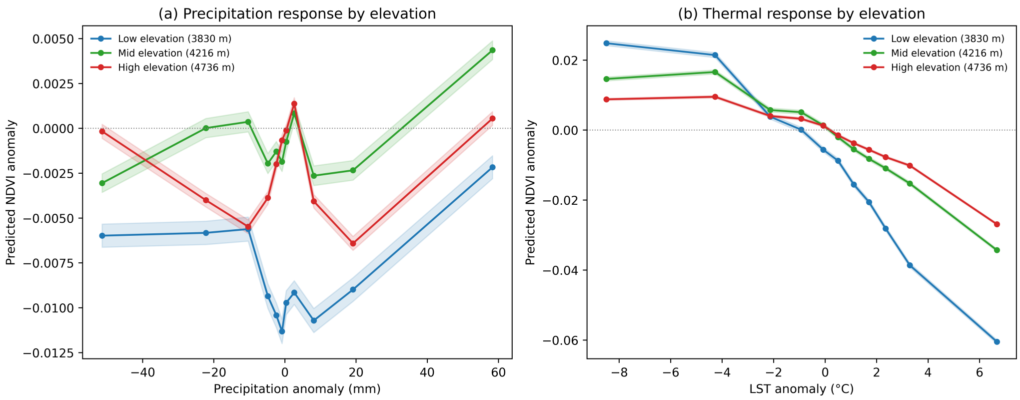

Figure 4: Conditional partial dependence of the interaction model I. Left: response of centered NDVI to temperature anomalies at three elevation percentiles; the thermal sensitivity decreases with elevation. Right: response to precipitation anomalies is weak and non-monotonic. Shaded areas denote ±1 standard error.

4.5 Pooled model C

The pooled model C, which combines geographic, topographic, seasonal and climate features on the raw (non-centered) NDVI, reaches $R^2 = 0.823 \pm 0.038$ under leave-one-year-out validation and $R^2 = 0.817 \pm 0.011$ under spatial block cross-validation, with RMSE of 0.054 and 0.056 respectively (Table 3). Two results stand out. First, the two validation schemes agree closely, indicating that the model generalizes across both time and space rather than exploiting leakage. Second, the pooled $R^2 = 0.82$ is far below the $R^2 = 0.98$ of the naive diagnostic, and the gap (≈ 0.16) quantifies precisely how much of the apparent skill of naive space–time models is an artifact of autocorrelation rather than predictive structure (Ogle et al. 2015). The importance profile (Table 4) is led by seasonality (0.285), latitude (0.244) and elevation (0.216), and the SHAP analysis of the pooled model assigns the largest mean absolute contributions to the seasonal harmonics (0.0515), latitude (0.0391) and elevation (0.0319), with SPI3 at 0.0162; the climate variables, although modest in absolute importance, are the only ones that carry interannual information beyond the seasonal cycle.

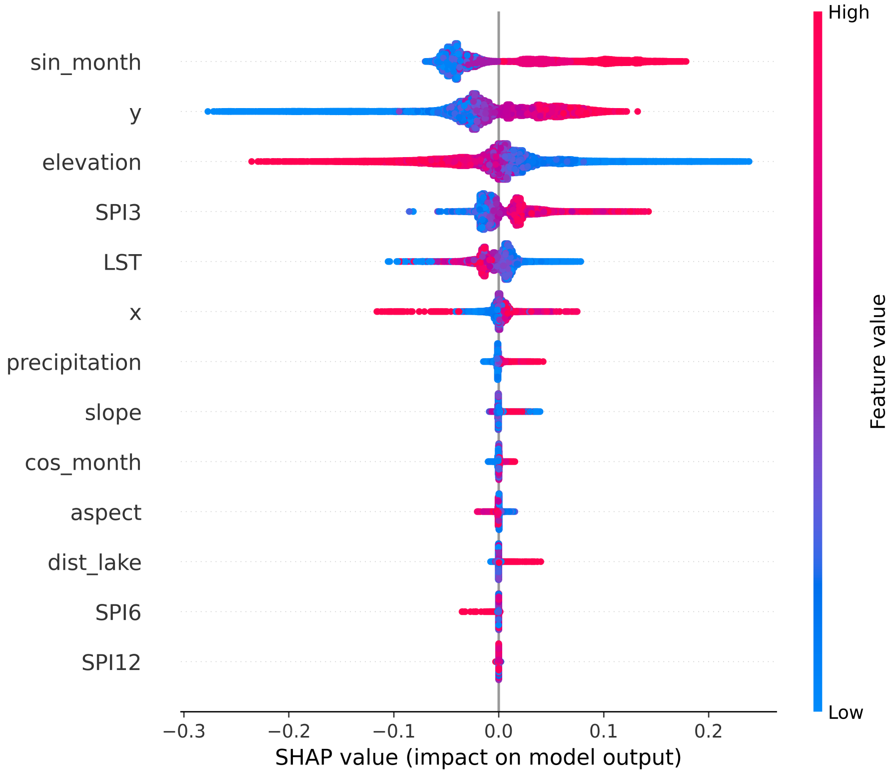

Figure 5: SHAP values of the pooled model C (leave-one-year-out). Seasonal harmonics, latitude and elevation dominate; climate variables contribute the interannual residual signal.

Table 3: Performance of all experiments (mean ± SD across folds). LOYO: leave-one-year-out; BlockCV: 10-fold spatial block cross-validation.

Experiment	Validation	R^2	RMSE	MAE
S (RF)	10-fold (pixels)	0.785 ± 0.041	0.055	—
S (OLS)	10-fold (pixels)	0.706 ± 0.040	—	—
T0	LOYO	0.544 ± 0.044	0.049	0.037
T1a	LOYO	0.691 ± 0.055	0.046	0.034
T1b	LOYO	0.517 ± 0.173	0.047	0.035
T1c	LOYO	0.649 ± 0.222	0.045	0.033
I	LOYO	0.750 ± 0.056	0.041	0.031
C	LOYO	0.823 ± 0.038	0.054	0.039
C	BlockCV	0.817 ± 0.011	0.056	0.041

Table 4: Impurity-based feature importance of the main random forests (top predictors).

Model	Top features (importance)
S	y (0.553), elevation (0.255), x (0.149), slope (0.016), aspect (0.016), dist_lake (0.010)
T1a	sin_month (0.842), LST_anom (0.093), precip_anom (0.052)
I	sin_month (0.786), elevation (0.083), LST_anom×elevation (0.039), LST_anom (0.038), precip_anom (0.019), precip_anom×elevation (0.018)
C	sin_month (0.285), y (0.244), elevation (0.216), x (0.069), LST (0.065), SPI3 (0.050), precipitation (0.018)

4.6 Synthesis

Figure 6 summarizes the experimental ladder. The monotonic increase from the seasonal baseline through the climate-anomaly and interaction models, and the convergence of the pooled model across two leakage-robust validation schemes, supports the central conclusion: the predictable signal of high-Andean vegetation is dominated by the spatial gradient ($\approx 70\%$ of variance, nearly all explainable by geography and topography), while the temporal signal is driven by seasonality and, to a lesser but robust extent, by temperature anomalies whose effect diminishes with elevation.

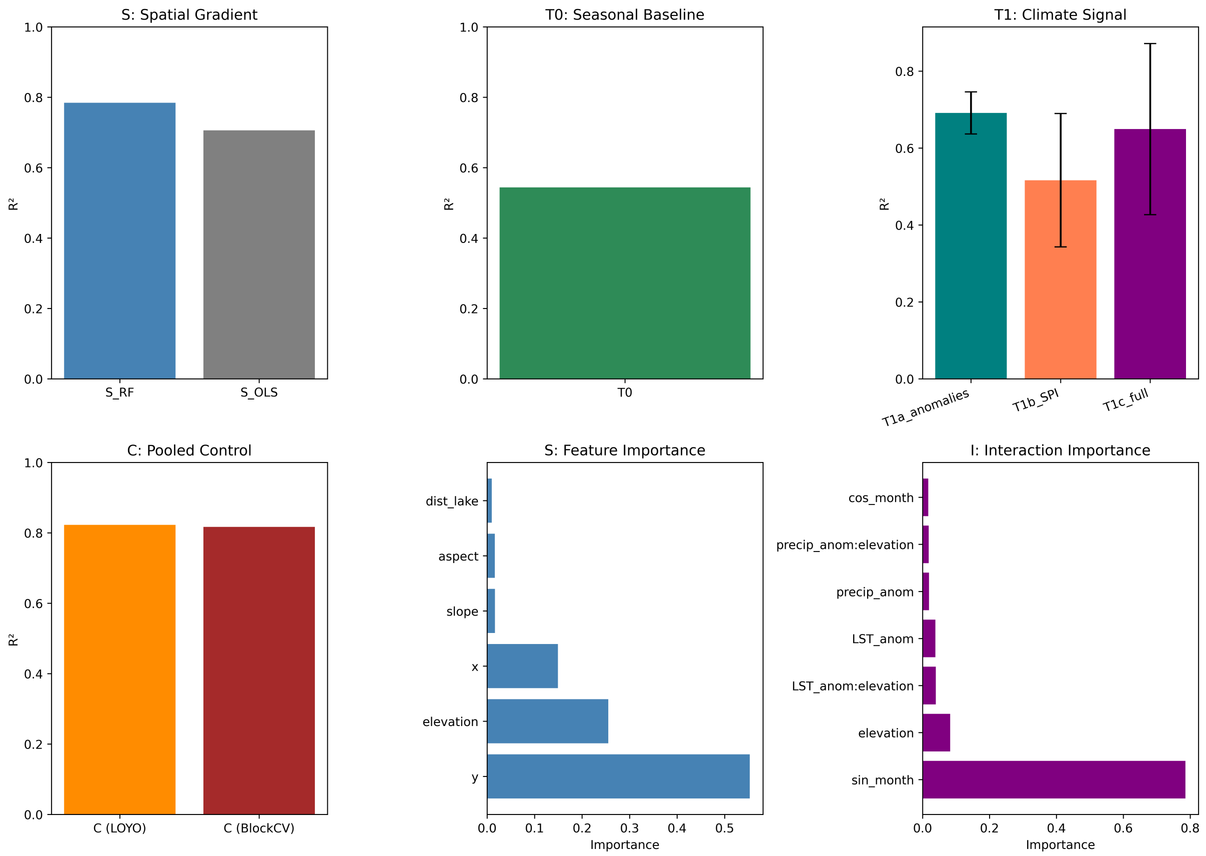

Figure 6: Synthesis of the experimental ladder: R^2 of each experiment under its validation scheme (top panels) and feature importances of the spatial and interaction models (bottom panels).

5 Discussion

5.1 The spatial gradient dominates high-Andean greenness

The finding that two-thirds of NDVI variance in the Peruvian Altiplano is spatial echoes the geomorphological reality of an ecosystem organized by elevation and by the regional moisture gradient (Garreaud 2009; Canedo-Rosso et al. 2018). Latitude, not elevation, is the leading single predictor, because the southeast–northwest precipitation asymmetry of the plateau produces greenness contrasts that exceed the thermal gradient. This has a methodological implication: pooled models that do not control for the spatial structure will systematically overestimate the role of climate, since a large share of the apparent "climate" response is in fact the spatial moisture gradient. The variance-partitioning framework makes this explicit and should be a standard first step in space–time vegetation studies (Borcard et al. 1992; Meyer, Reudenbach, et al. 2019).

5.2 Temperature anomalies, not drought, drive interannual dynamics

Within the temporal component, the climate signal beyond seasonality is dominated by temperature anomalies: T1a outperforms the seasonal baseline by almost 15 percentage points, and the SPI-based models fail to add information (T1b $R^2 <$ T0). Two mechanisms explain the thermal dominance. First, at altitudes where vegetation is already water-limited but not fully temperature-limited, positive thermal anomalies increase evaporative demand and accelerate senescence; the steepening of the negative response at lower elevations (Figure 4) is consistent with this mechanism. Second, the hydrological signal is attenuated by the monthly aggregation of a strongly convective precipitation regime and by the acknowledged uncertainty of CHIRPS where gauge networks are sparse, as in the southern highlands (Canedo-Rosso et al. 2018); the negative SPI result discussed in Section 4.3 is consistent with this attenuation. A further caveat is that LST and NDVI are retrieved from the same MODIS platform, so part of the LST–NDVI coupling may reflect cloud radiative effects (cold, cloudy scenes suppress both) rather than plant physiology alone; separating the two would require independent thermal forcing such as reanalysis or other sensors.

5.3 Elevation modulates the climate response

The significant $LST\_anom \times elevation$ interaction and the partial-dependence profiles indicate that thermal sensitivity decreases with elevation. At 3830 m, near the lower boundary of the puna, vegetation is dense enough to respond strongly to heat stress; above 4700 m, sparse and temperature-limited vegetation buffers the response. This elevational gradient of sensitivity is a specific, testable prediction of the analysis and aligns with the elevation-dependent climate change documented in Andean regions (Pepin et al. 2015; Vuille et al. 2015): if warming intensifies, the low-elevation fringe of the puna is the most vulnerable zone.

5.4 Robust pooled performance without leakage

The close agreement between LOYO and BlockCV ($R^2 = 0.82$ vs. 0.82) and the gap with the naive diagnostic ($R^2 = 0.98$) is the paper's central methodological result. It shows that honest space–time evaluation is achievable with standard tools (Roberts et al. 2017; Meyer and Pebesma 2022) and that the previous inflation was almost entirely attributable to temporal autocorrelation and the spatial gradient rather than to real predictive structure. It also reconciles our results with studies that report very high NDVI predictability from lagged or autocorrelated designs (Ogle et al. 2015): those designs measure persistence, not climate response.

5.5 Limitations

Six limitations temper the conclusions. (1) The monthly temporal resolution and the 10-year window limit the detection of slow vegetation change and of fast hydrological responses. (2) CHIRPS precipitation, used both as a predictor and as the basis of the SPI, is uncertain where gauge networks are sparse, and the SPI itself is precipitation-only; a product that integrates temperature or evapotranspiration (e.g., SPEI, or reanalysis such as ERA5-Land, or the gauge-based PISCO) could alter the hydric results. (3) The MODIS-based compositing can retain scan-angle artifacts in the 1 km product, propagated through the resampling of coarser climate grids to the analysis raster; the study-area fields examined here show no detectable high-frequency banding, but Fourier-domain destriping filters or the Harmonized Landsat–Sentinel (HLS) data would remove any residual orbital track noise in future applications. (4) Impurity-based importance is biased toward continuous and high-cardinality features, which is why we corroborated the main models with SHAP; nonetheless, the ranking of the first-order predictors is robust across both metrics. (5) The variance partition is global, over a domain that includes strong internal gradients; decomposing it per elevation belt or per watershed is a natural extension. (6) The partial dependence of the precipitation response is locally unstable near zero anomalies because of the concentration of dry-season structural zeros; monotonic constraints or spatial generalized additive models would smooth the profile and are a natural methodological refinement.

6 Conclusions

This study provides the first explicit variance partition of high-Andean vegetation dynamics in the Peruvian Altiplano using machine learning and leakage-robust validation. The total NDVI variance is composed of 69.8% spatial (between-pixel) and 30.5% temporal (within-pixel) components. The spatial component is explained almost entirely by latitude, elevation and longitude ($R^2 = 0.785$), with latitude capturing the regional moisture gradient. The temporal component is dominated by seasonality ($R^2 = 0.544$), to which temperature anomalies add 14.7 percentage points ($R^2 = 0.691$), temperature–elevation interactions a further 5.9 points ($R^2 = 0.750$), and whose thermal sensitivity decreases monotonically with elevation. A pooled model reaches $R^2 = 0.823$ under leave-one-year-out and $R^2 = 0.817$ under spatial block cross-validation, directly exposing the autocorrelation inflation of naive space–time models. The elevation-modulated thermal response, and the primacy of the spatial gradient over interannual climate in total variability, are the main substantive findings, with direct implications for the vulnerability of the low-elevation puna fringe under accelerated Andean warming.

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Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Noe Ulises Machaca Chambilla: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Data availability

The aggregated dataset and the full analysis pipeline are available from the corresponding author on reasonable request. All source data are publicly available: MODIS MOD13Q1 and MOD11A2 (NASA, via Google Earth Engine), CHIRPS v2.0 precipitation (UCSB Climate Hazards Center), and SRTM v3 terrain (NASA/USGS).

Declaration of generative AI and AI–assisted technologies in the writing process

During the preparation of this work the author used large-language-model tools for drafting and language-editing assistance. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. All figures are data visualizations generated from reproducible analysis code and comply with the journal's policy on AI-generated imagery.

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A The historical pooled diagnostic

The initial pooled space–time random forest, fitted on the raw NDVI with a lag-1 autoregressive term and evaluated with a random split, achieved $R^2 = 0.98$ and left residuals with lag-1 autocorrelation of 95.4%. The model was dominated by the spatial gradient and by the persistence of the series, not by climate predictors: feature importance ranked the lagged NDVI first and latitude/elevation second, while the climate features ranked last. This diagnostic motivated the variance-partitioning framework and the leakage-robust validation used in the present study. Figure 7 shows the observed-versus-predicted plot and the importance profile of the naive model, and Figure 8 the temporal autocorrelation of the series that sustains the inflation.

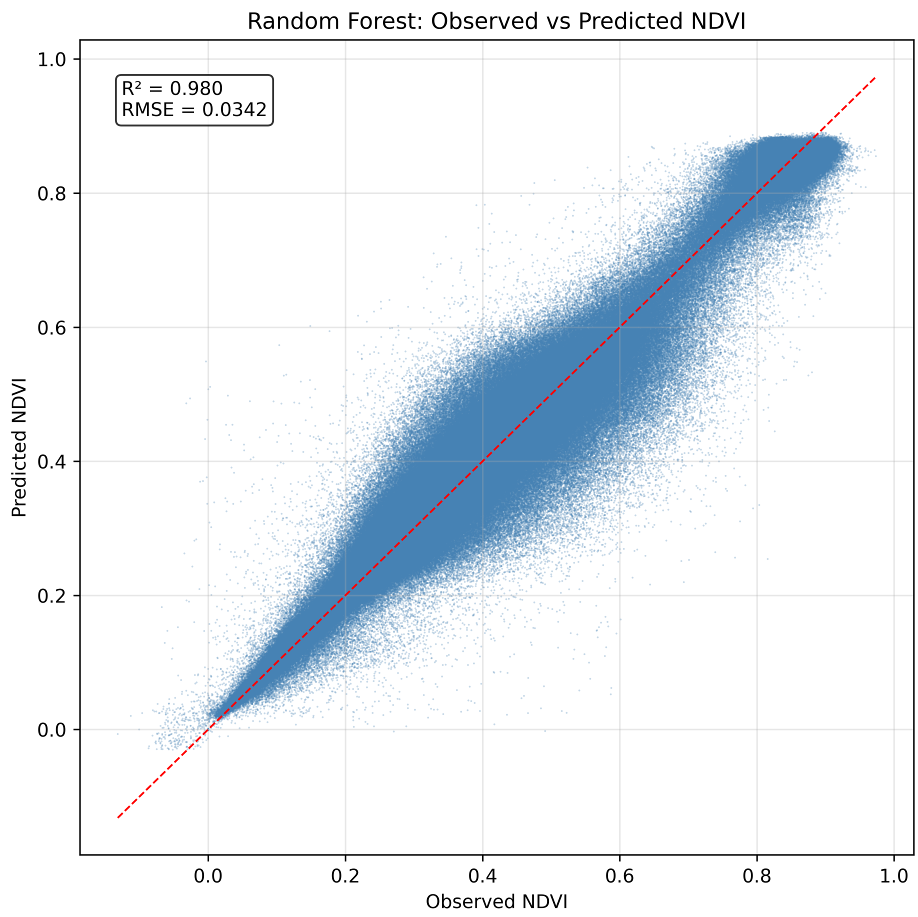

Figure 7: Naive pooled model (historical diagnostic): observed vs. predicted NDVI ($R^2 = 0.98$), inflated by spatial gradients and temporal autocorrelation.

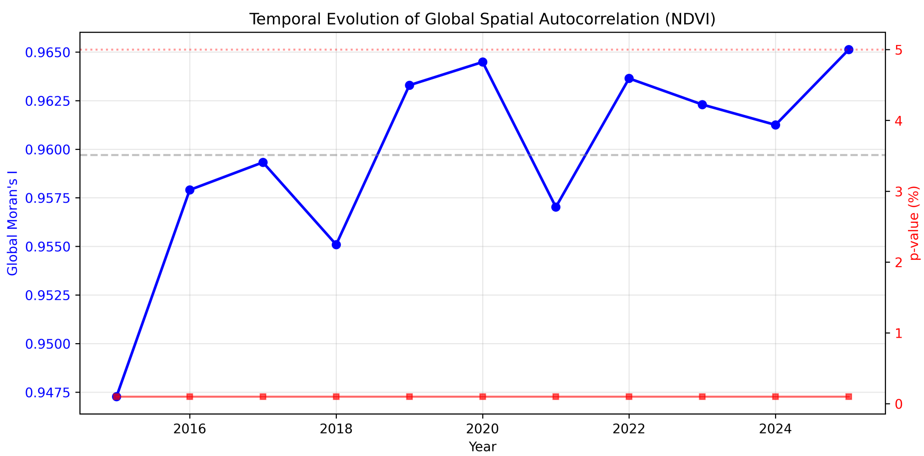

Figure 8: Temporal autocorrelation structure of the pooled NDVI series (historical diagnostic).